

BUILD-EMS

Behind-the-Meter Control for Rooftop Solar, Wind, Storage & EV — Multi-Tenant Ready

[Product image: BUILD-EMS / NESL 886B board]

Deploying EV charge/discharge with rooftop solar PV and wind on an existing building? **BUILD-EMS** is the open-source behind-the-meter controller that integrates **bidirectional DC/AC inverters**, multi-tenant “plug-in solar”, and EV infrastructure into one balanced building minigrid — enforcing multi-zone energy policy while protecting the mains.

What Sets BUILD-EMS Apart:

- ▶ **Bidirectional & Inverter-Ready** : coordinates DC/AC bidirectional inverters and converters for solar PV, wind and battery, with EV charge and discharge (V2X).
- ▶ **Multi-Tenant “Plug-in Solar”** : partitions shared rooftop generation and storage across tenants with per-zone, time-of-day energy policies.
- ▶ **Mains-Protective** : balances the building mains and responds to grid-peak DERMS events without exceeding the service limit.
- ▶ **Open-Source Hardware** : released under CERN-OHL-S v2 — schematics, board files and firmware published on GitHub (meshEMS / X-EMS).
- ▶ **Field-Serviceable** : built from open, off-the-shelf commodity modules, snap-in mounted for fast in-country assembly and repair.

BUILD-EMS turns a building rooftop into a coordinated, multi-tenant minigrid — maximizing solar, wind and EV flexibility while keeping the mains within limits.

Intelligent

- ▶ Multi-zone, time-of-day energy policy
- ▶ Mains balancing & DERMS peak response
- ▶ EV charge / discharge (V2X) scheduling

Interoperable

- ▶ Bidirectional DC/AC inverters (PV / wind / battery)
- ▶ SunSpec DER + Modbus / CAN
- ▶ Multitenant sub-metering & EVSE

Resilient

- ▶ Local policy enforcement, cloud-optional
- ▶ Per-tenant energy allocation
- ▶ Open meshEMS platform

PRODUCT	
Product name	BUILD-EMS (10Power X-EMS, supplied by NESL)
Model / PCB	NESL EMS Controller PCB 886B (shared X-EMS platform)
Firmware	meshEMS / OpenAMI metering (open source, ~50k LOC middleware)
License & source	Hardware: CERN-OHL-S v2 Repository: GitHub, meshEMS / X-EMS
Serviceability	Open, off-the-shelf commodity modules snap-in mounted for fast in-country assembly and repair
COMPUTE	
Processor	Dual-core 32-bit MCU @ 240 MHz WiFi + Bluetooth LE
Memory	16 MB Flash + 8 MB PSRAM
ENERGY MANAGEMENT	
Supported DER	Solar PV wind battery, via bidirectional DC/AC inverters / converters
EV infrastructure	EVSE charge & discharge (V2X) per-tenant charging needs
Energy policy	Multi-zone time-of-day policies per-channel kW / kWh limits multitenant allocation DERMS response
ASSET COMMUNICATION	
Serial buses	2 × RS-485 Modbus: EV-charger (EVC) + sensor/DTM ports 1 × CAN 2.0
On-board metering	SPI poly-phase metering, up to 6 CT channels bidirectional (import/export)
Local I/O	2 × onboard SSR outputs door-switch input 2 analog buttons
SUPPORTED MODULES & SENSORS (out of the box)	
DER & inverters	SunSpec models 1 / 11 / 213 (bidirectional inverters & meters)
Energy meters / relay	Single-phase Modbus meters (bidirectional kWh) 8-ch I ² C SSR bank
Sensors / networking	Temp / humidity sensor Ethernet WiFi / BLE
POWER & ENCLOSURE	
Operating voltage	5 V DC — USB-C or screw terminals or 4.5–20 V DC or 8–12 V AC (SELV)
Power / dimensions	85 × 55 mm or Raspberry Pi HAT form factor
STANDARDS & POSITIONING	
Grid position	Behind-the-meter (BTM) building / commercial rooftop minigrid
Capabilities	EV V2X multitenant solar / wind / storage mains balancing & DERMS

Customizable with added hardware modules and open or licensed code stacks. BUILD-EMS is an open-source X-EMS product supplied by New Energy Solutions Lab for 10Power.